25/2/26 Exp. No:- 6

Verification of maximum power Transfer theorem:-

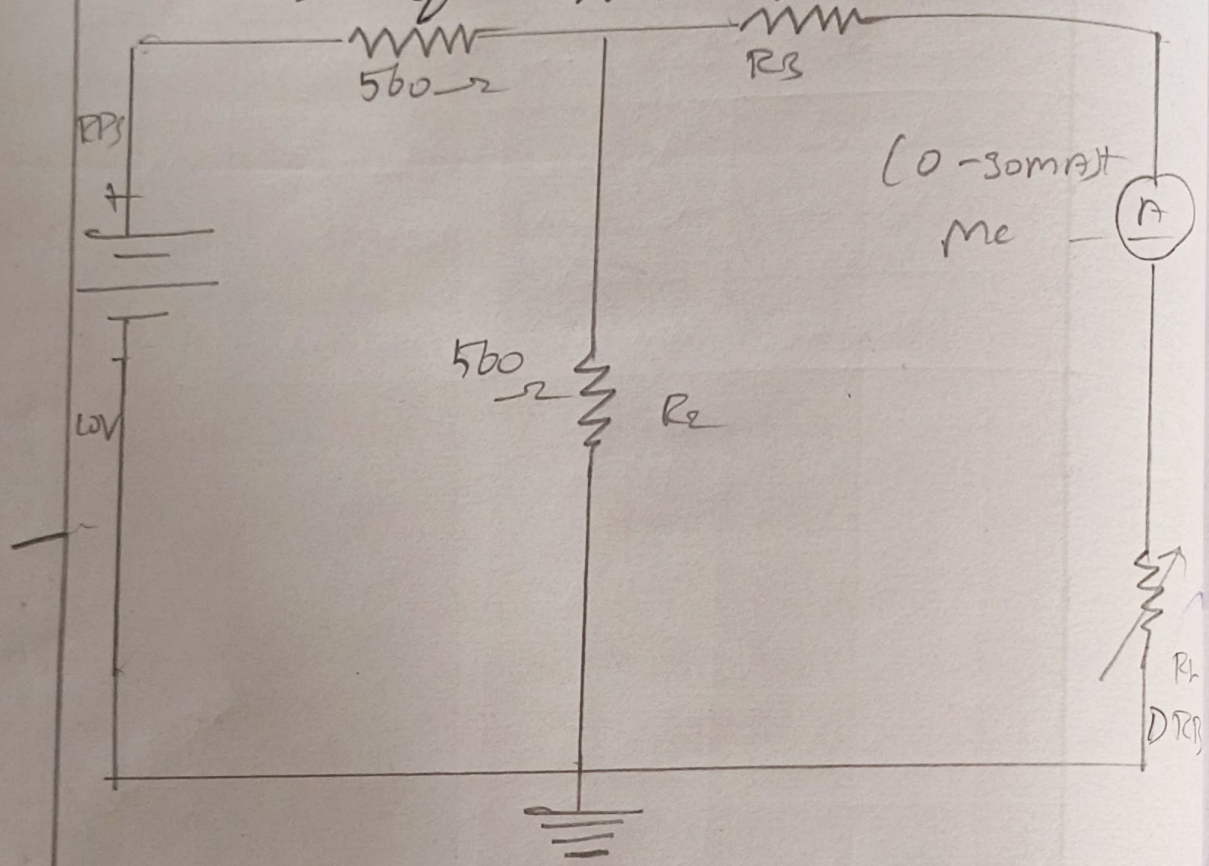
Aim:-

To measure the power absorbed in a load and to verify that the power absorbed in a load is maximum only when load resistance is equal to the source resistance.

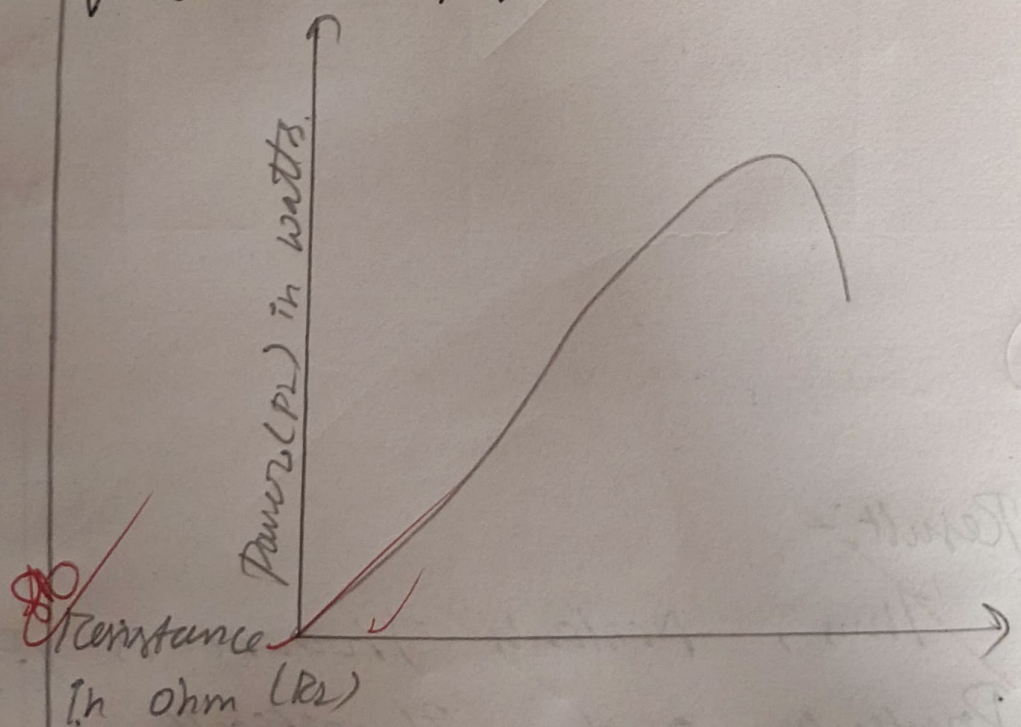
Apparatus Required:-

S.NO	Name of the apparatus	Range / Rating	Quantity
1	Voltmeter	(0-15V) MC	1
2	Ammeter	(0-500mA) MC	1
3	Resistors	560-Ω 470-Ω.	2, 1.
4	RPS (DC Supply)	15V	1

Circuit diagram: For maximum power Transfer Theorem:-



Model Graph:-



Tabulation:-

Table (1): For Theoretical Calculation: $R_{th} = 750\Omega$.

S.NO	Load Resistance (R_L) in ohm	Load current (I_L) in mA	Load voltage (V_L) in volts	Load power (P_L) in watts
1	11k	6.84	10	68.4
2	47	6.82	10	68.2
3	1m	6.83	10	68.3
4	15k	6.78	10	67.9

O/P verified

Procedure:-

1) Make connection as per the circuit diagram.

2). Change the resistors R_L whose value close to R_{th}, measure the corresponding V_L, I_L and calculate the power P_L and enter into the table (2).

3). plot a graph between R_L and P_L and find the R_L corresponding to maximum power transfer.

4). Verify the measured values of R_L at maximum power transfer as same as calculated and found graphically.

Table (2): For Practical Calculation:-

S.NO	Load Resistance (R_L) in ohm	Load Current (I_L) in amps	Load Voltage (V_L) in volts	Load power (P_L) in watts.
1	1K	6.85	10	68.5
2	47	6.83	10	68.3
3	1M	6.84	10	68.4
4	15 K	6.79	10	67.9.

Model calculation:-

Given:

$R_1 = 560\Omega$

$R_2 = 560\Omega$

$R_3 = 470\Omega$

Supply voltage = 10V

Step 1: Find R_{th}

$\Rightarrow R_{th} = R_3 + \left(\frac{1}{R_1} + \frac{1}{R_2}\right)$

Marks split-up:

S.No	Description	Marks allotted	Marks obtained.
1	Experimental setup / connections	20 marks	20
2	Observation / Readings	20 marks	20
3	Calculations / Results	20 marks	20
4	Viva	20 marks	20
5	Record	20 marks	20
		Total.	100

$$= 470 + \left(\frac{560 \times 560}{560 + 560}\right)$$

$$= 470 + 280$$

$$= 750\ \Omega$$

Step 2: find V_{th}

$$V_{th} = 10 \times \frac{R_2}{R_1 + R_2} = \frac{10 \times 560}{560 + 560}$$

$$= 5V$$

Step 3: $R_L = 750\ \Omega$

$$R_L = R_{th}$$

$$R_L = 750$$

$$I_L = \frac{V_{th}}{R_{th} + R_L} = \frac{5}{760 + 760} =$$

$$= 0.0032\ A$$

$$\Rightarrow V_L = I_L \times R_L$$

$$= 2.5V$$

$$\Rightarrow R_L = V_L \div I_L$$

$$= 8.33\ \Omega$$

10
19/3

Result:-

Thus, maximum power transfer theorem is verified. Practically and theoretically.